

Left-closed Sets

§ Left-closed

Parameters

- + $A \in \text{Set}$
- + $R \in A \sim A$
- + $X \in \text{Set}$

Assumptions

- a. $X \subseteq A$
- b. $(\forall a, b \mid a R b \wedge b \in X : a \in X)$

Observations

1. $X = R[X] \Rightarrow \text{Left-closed } R \ X$
2. $X = R[X] \Leftarrow \text{Left-closed } R \ X \wedge \text{Reflexive } R$
3. $(\text{Left-closed } R \ X \equiv X = R[X]) \Leftarrow \text{Reflexive } R$

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Proof of Observation 1.

- ◆ $X = R[X]$
- ≡ Extensionality
- $(\forall a \mid : a \in X \equiv a \in R[X])$
- ⇒ Predicate calculus
- $(\forall a \mid a \in R[X] : a \in X)$
- ≡ Direct image
- $(\forall a \mid (\exists x \mid x \in X : a R x) : a \in X)$
- ≡ Splitting
- $(\forall x \mid x \in X : (\forall a \mid a R x : a \in X))$
- ≡ Nesting
- $(\forall a, x \mid x \in X \wedge a R x : a \in X)$
- ≡ Definition of Left closed
- Left closed $R \ X$
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Proof of Observation 2.

◆ $X = R[X]$
 ≡ Extensionality
 $(\forall a \mid : a \in X \equiv a \in R[X])$
 ≡ Mutual implication
 $(\forall a \mid a \in X : a \in R[X]) \wedge$
 $(\forall a \mid a \in R[X] : a \in X)$
 ≡ Definition of Left closed
 $(\forall a \mid a \in X : a \in R[X]) \wedge$
 Left set $R \ X$
 ≡ Identity of $\wedge$
 ◆ Left closed $R \ X$
 ... $(\forall a \mid a \in X : a \in R[X])$
 ≡ Definition of direct image
 $(\forall a \mid a \in X : (\exists x \mid x \in X : a \ R \ x))$
 ≡ Reflexivity of R , using assumption a.
 ◆ Reflexive R
 ... true
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Proof of Observation 3.

◆ $X = R[X]$
 i. Reflexive R
 $\Rightarrow$ Observation 1.
 Left-closed $R \ X$
 ≡ Assumption i.
 Left-closed $R \ X \wedge$ Reflexive R
 $\Rightarrow$ Observation 2.
 $X = R[X]$
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